

Supplementary Information

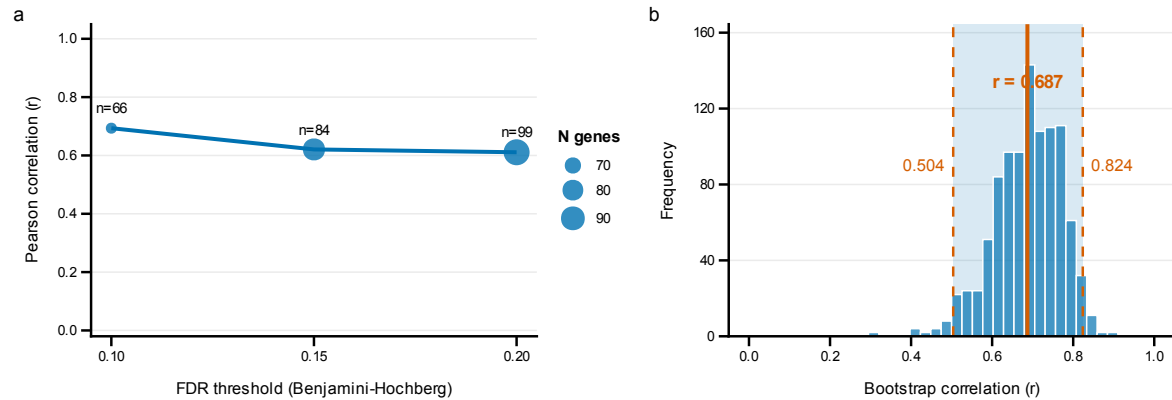
A Multi-Tissue Transcriptomic Atlas of Porcine Development Identifies Conserved Molecular Programs with Humans

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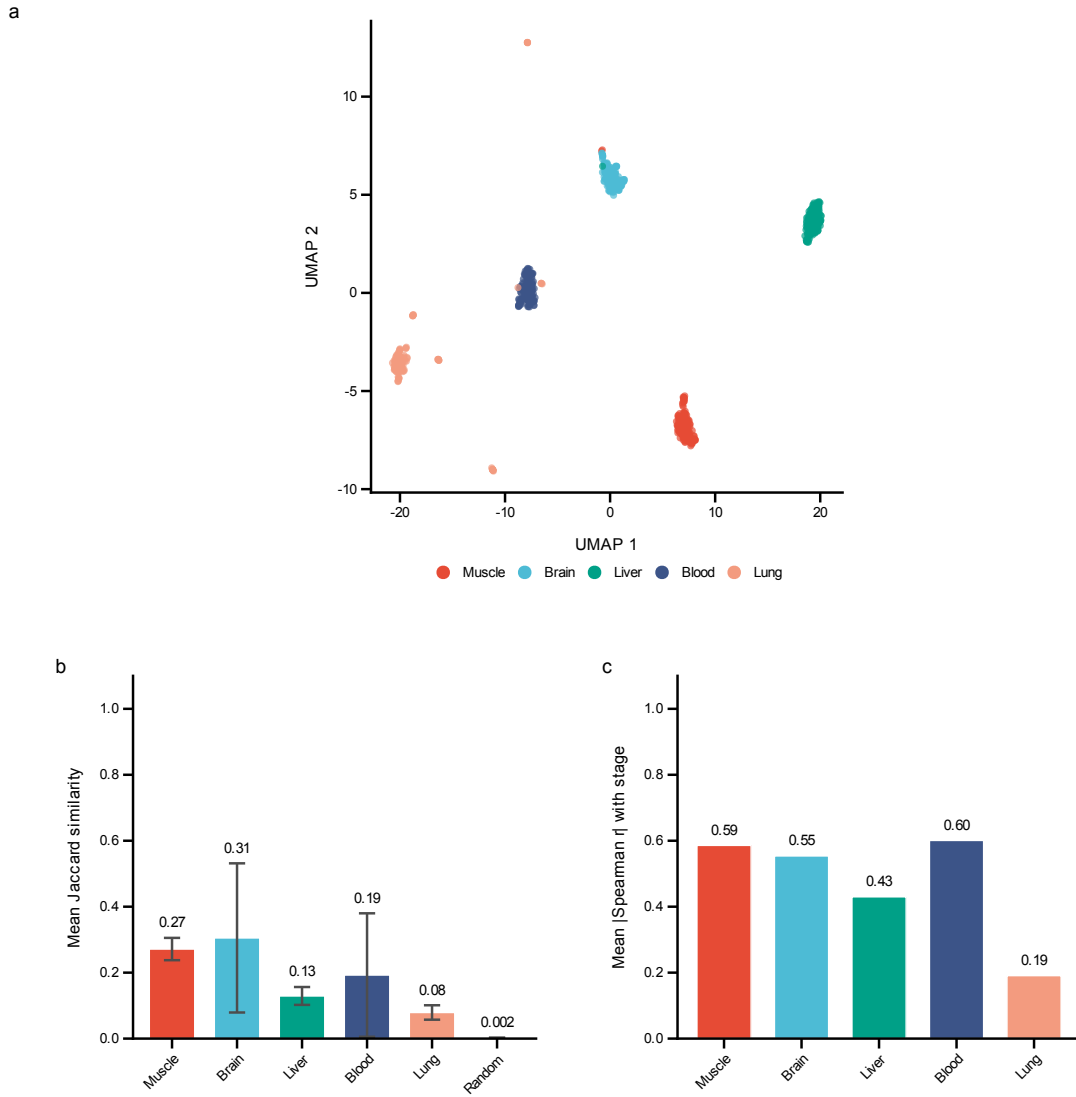
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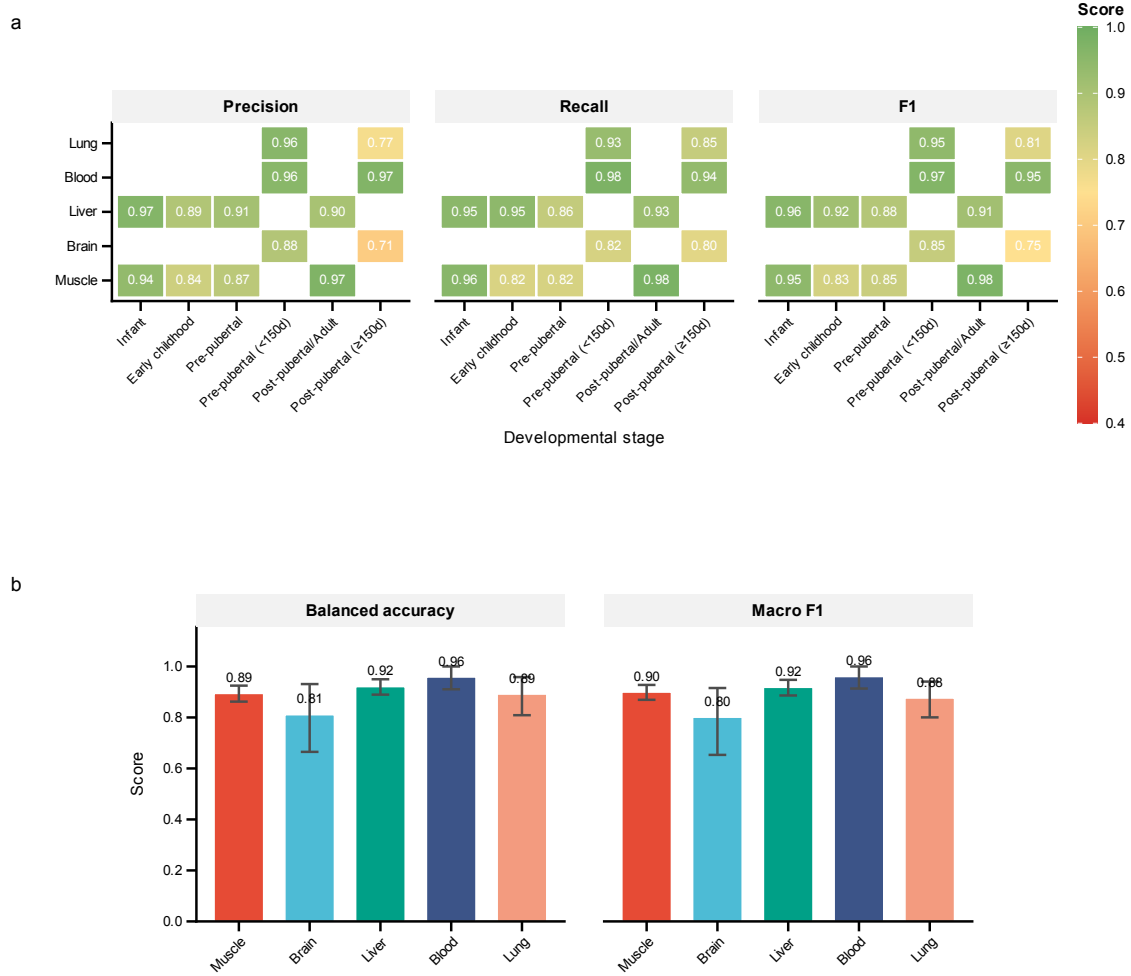
Supplementary Figures



Supplementary Figure 1: Cross-species sensitivity analysis of conserved developmental markers. **a**, Pearson correlation between pig and human developmental fold changes across FDR thresholds ranging from < 0.05 to < 0.20 (Benjamini–Hochberg corrected), using all genes passing pre-specified criteria (FDR $<$ threshold and $|\log_2 \text{FC}| > 0.5$ in both species). Points are sized by the number of genes at each threshold. **b**, Bootstrap distribution of correlation coefficients ($n = 1,000$ iterations) for the pre-specified criteria (FDR < 0.10 , $|\log_2 \text{FC}| > 0.5$, $n = 66$ genes). Solid vertical line indicates the observed correlation ($r = 0.693$); dashed lines denote the 95% bootstrap confidence interval boundaries (0.50–0.82).

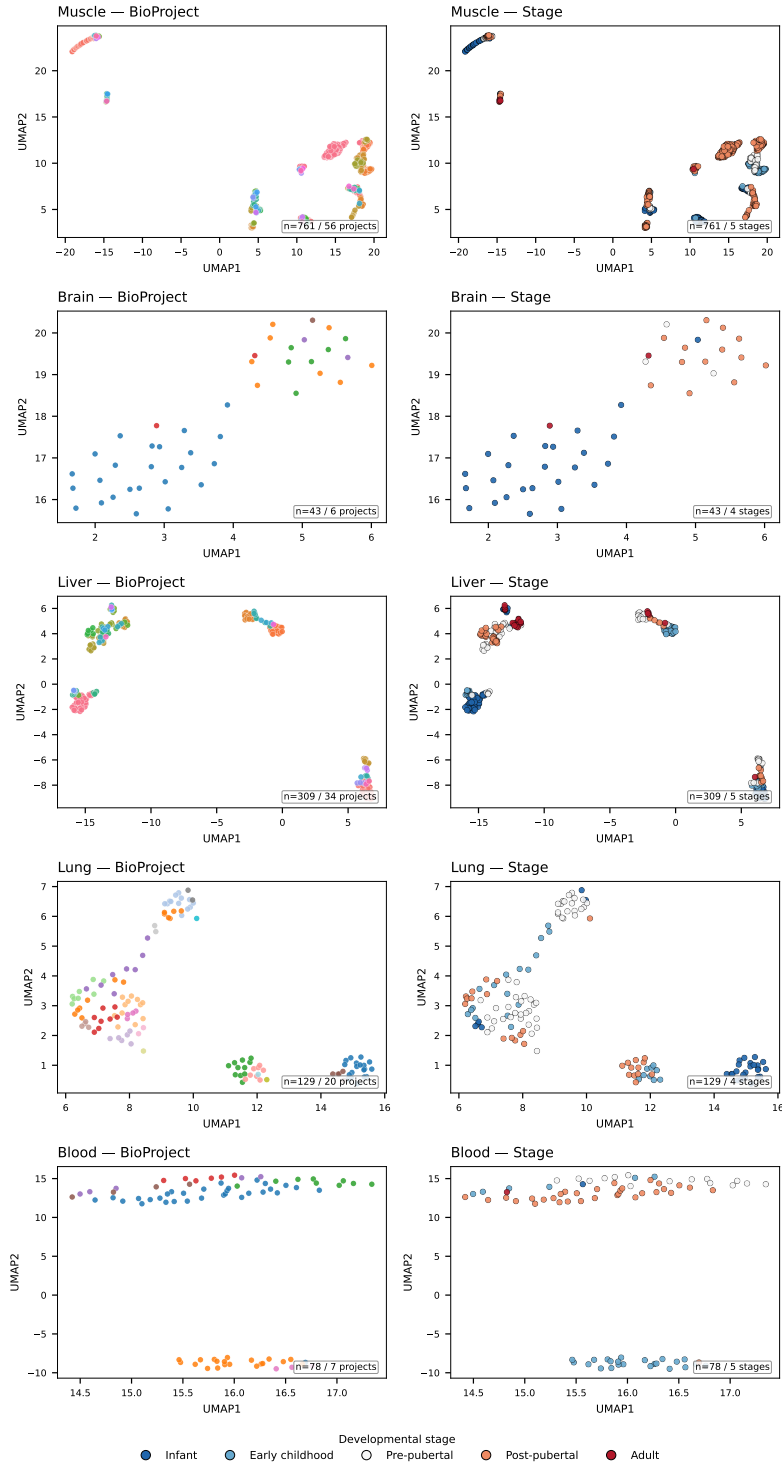


Supplementary Figure 2: Tissue specificity and feature selection stability analysis. **a**, Uniform manifold approximation and projection (UMAP) of gene expression profiles across five tissues (muscle, brain, liver, blood, lung), demonstrating distinct tissue-specific clustering patterns ($n = 150$ samples per tissue, balanced sampling; top 500 variable genes). For visualization only, BioProject batch effects were removed per tissue using `limma::removeBatchEffect` prior to UMAP; the machine learning pipeline operates on uncorrected expression data. **b**, Within-tissue feature selection stability quantified as mean Jaccard similarity index of the top 50 selected features across ten independent random seeds. Error bars represent standard deviation; higher values indicate more reproducible feature selection. **c**, Mean absolute Spearman correlation between top-ranked features and developmental stage for each tissue.

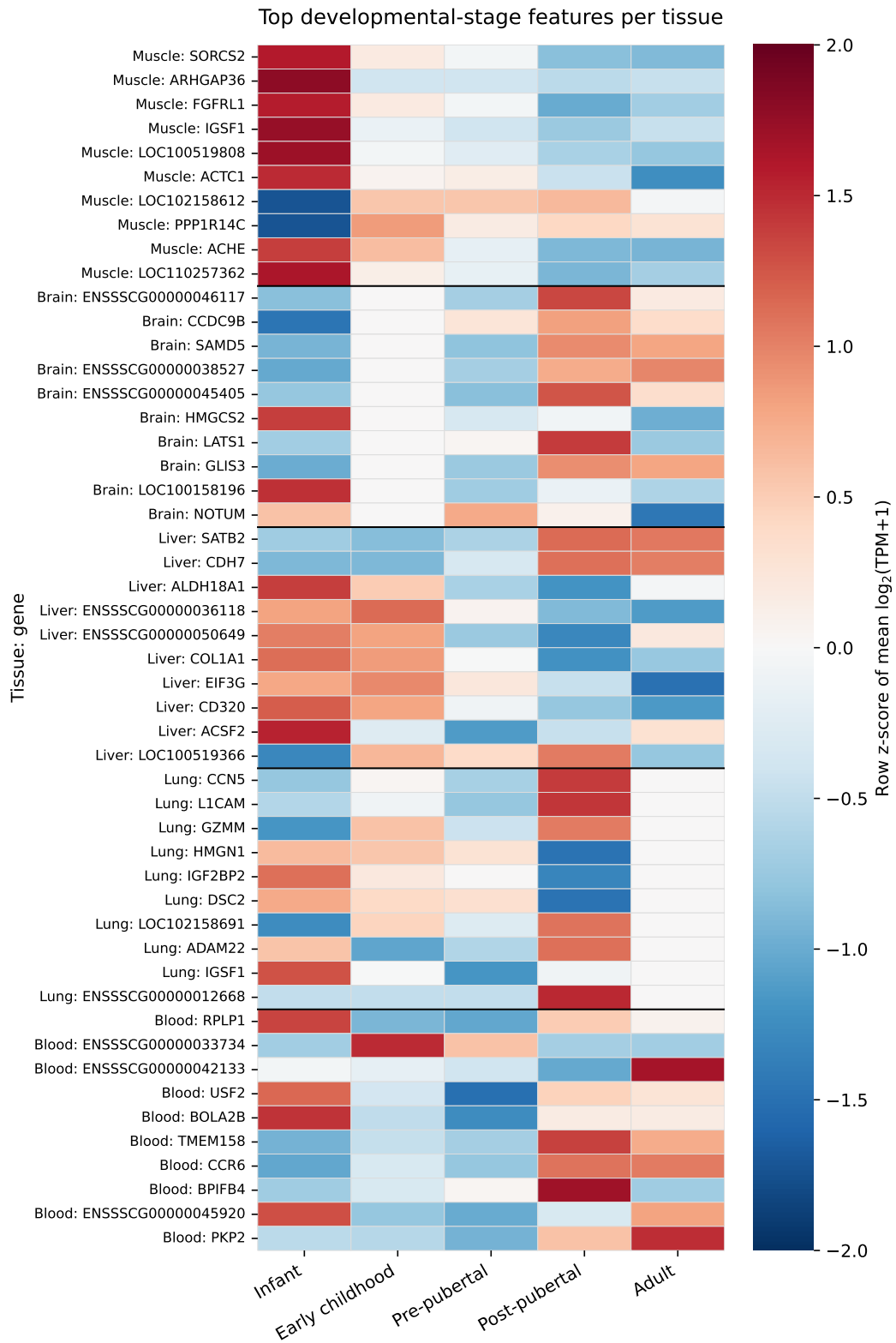


Supplementary Figure 3: Extended classification performance metrics across tissues and developmental stages. **a**, Heatmap of per-class precision, recall, and F1 scores for each developmental stage across all five tissues. Colour scale ranges from 0.4 (red, poor performance) to 1.0 (green, excellent performance). Missing values indicate developmental stages not present in that tissue's classification scheme. **b**, Overall performance summary showing balanced accuracy and macro-averaged F1 scores evaluated on pooled cross-validated held-out predictions for each tissue. Error bars indicate 95% bootstrap confidence intervals from 1,000 resampling iterations.

Batch-effect evaluation across five tissues: developmental stage drives the dominant variance even when samples span multiple BioProjects.

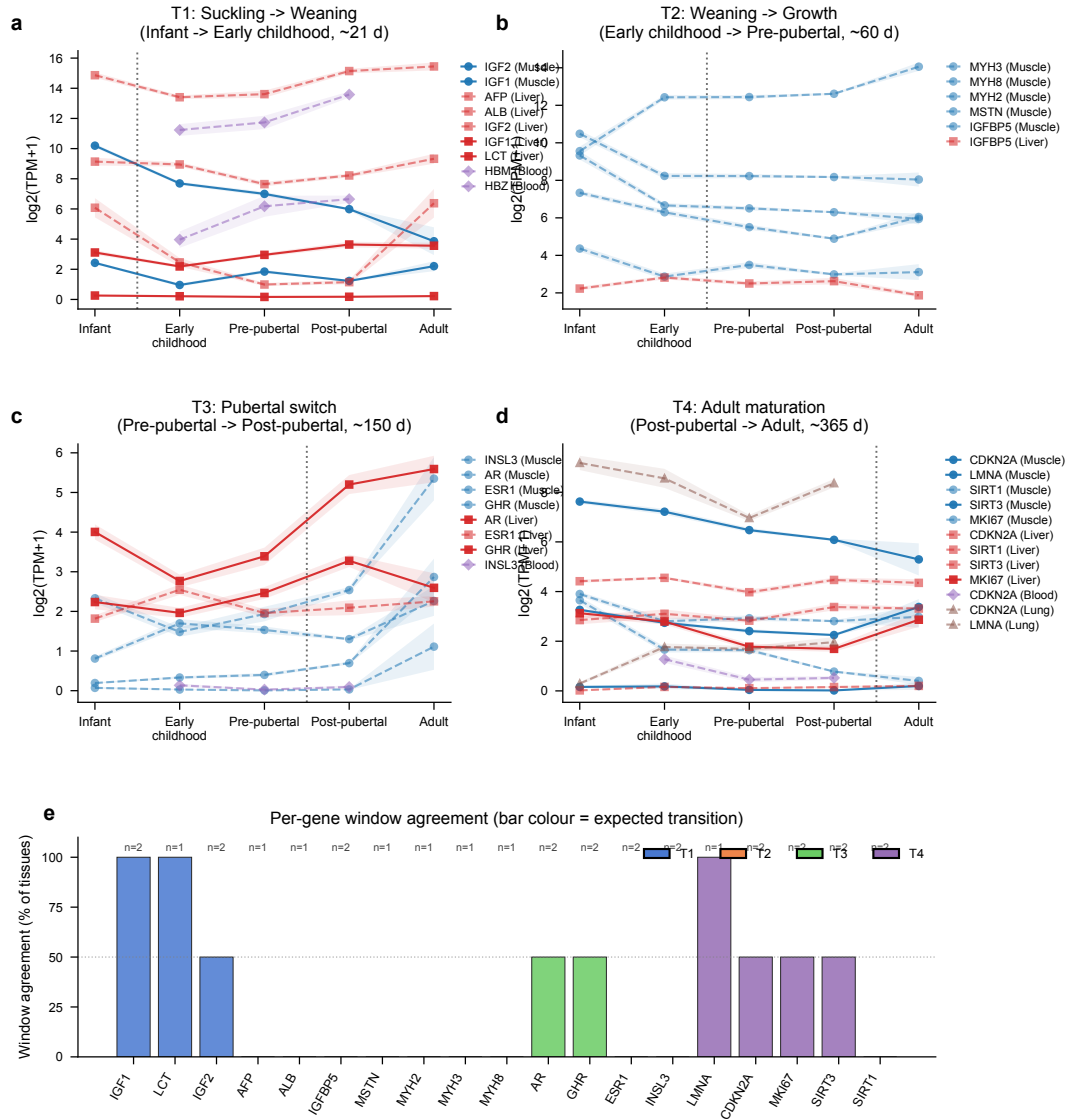


Supplementary Figure 4: Batch-effect evaluation across five tissues: developmental stage drives visible variance even when samples span multiple BioProjects. Per-tissue uniform manifold approximation and projection (UMAP) embeddings of the top 5,000 most-variable genes ($\log_2(\text{TPM} + 1)$), one row per tissue (muscle, brain, liver, lung, blood). **Left column**, points coloured by source BioProject (categorical palette). **Right column**, points coloured by developmental stage (ordered palette infant $\rightarrow$ adult). Sample counts (n) and the number of distinct BioProjects spanning each tissue are annotated. UMAP parameters: `n_neighbors = 15`, `min_dist = 0.1`, `metric = "euclidean"`, `random_state = 42`; PCA used for upstream variance decomposition (10 components). For most tissues, infant-to-adult colour gradients are visible even where several BioProjects co-occupy the embedding, supporting the interpretation that the dominant variance components are not solely technical. Liver shows the strongest BioProject structuring; its developmental component is recovered at PC5 (see Supplementary Table S6 for the full variance decomposition).



Supplementary Figure 5: Top developmental-stage features per tissue with stage-wise expression trajectories. Heatmap of row-z-scored mean $\log_2(\text{TPM} + 1)$ values across the five developmental stages (infant $\rightarrow$ adult) for the top 10 features per tissue, ranked by aggregate LightGBM importance across the $K - 1$ binary classifiers within each tissue's calibrated ordinal model. Each panel/section corresponds to one tissue (muscle, brain, liver, lung, blood). Row order follows model importance (top to bottom = most to least important). The heatmap surfaces the timing of each gene's peak expression and the magnitude of its developmental trajectory; functional annotations for these genes are in Supplementary Table 8.

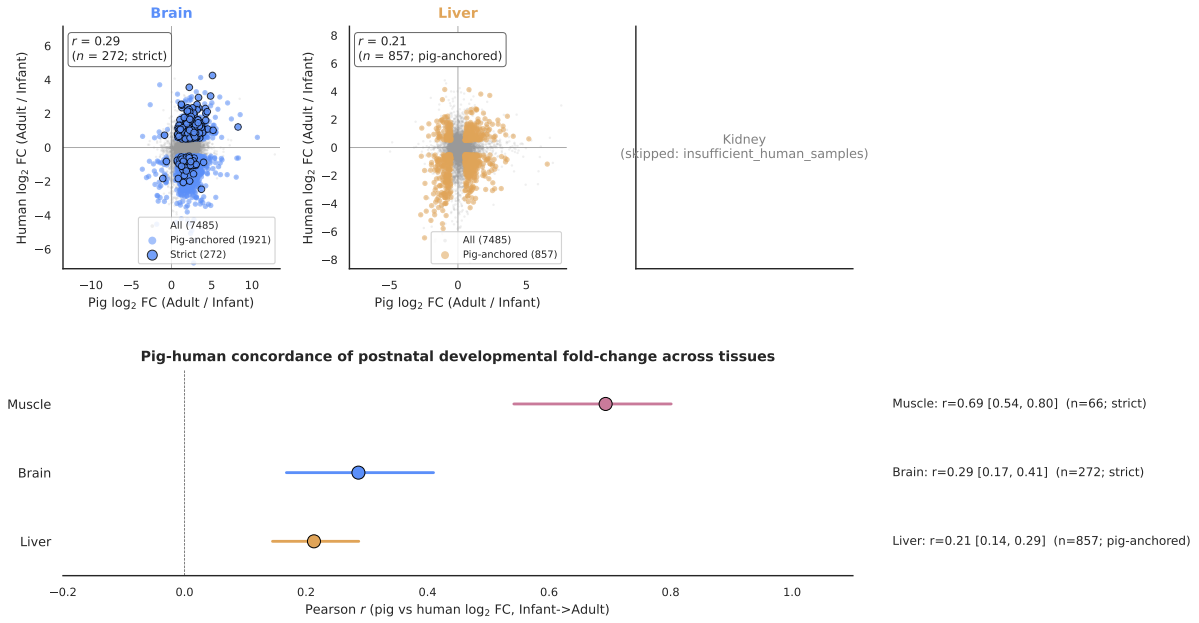
Figure S6. Biological validation of porcine developmental stages via curated marker genes



Lines: mean $\log_2(\text{TPM}+1)$; shaded band: ± 1 SE across samples. Vertical dotted line: expected literature-defined transition. Solid line = observed max-change window matches expected; dashed = mismatch.

Supplementary Figure 6: Independent biological validation of stage boundaries using a curated panel of 25 literature-supported postnatal marker genes. Stage-wise $\log_2(\text{TPM} + 1)$ trajectories (line with shaded SE band) for canonical markers of the four transition windows: T1 (suckling-to-weaning ~21 d), T2 (weaning-to-growth ~60 d), T3 (pubertal switch ~150 d), and T4 (adult maturation ~365 d). The vertical dashed line in each panel marks the literature-predicted transition. The final panel shows the per-gene observed-vs-expected window agreement (bar height indicates whether the gene's largest adjacent-stage change occurred at the predicted window). Of 30 marker $\times$ tissue pairs with adequate per-stage coverage (≥ 5 samples per flanking stage), 20/30 (67%) matched the predicted direction of change and 10/30 (33%) had their largest single-step change at the predicted transition; T1 (suckling $\rightarrow$ weaning) had the highest window-match rate ($4/7 = 57\%$). The porcine myosin heavy-chain switch (*MYH3*, *MYH8*, *MYH2*) was directionally correct but transitioned earlier than the assumed weaning boundary, consistent with the first-month perinatal isoform exchange documented in pig skeletal muscle. Full gene-level results are in [review/analyses/results/marker_gene_transitions.csv](#).

Figure S7. Cross-species concordance of developmental fold-change extends beyond skeletal muscle

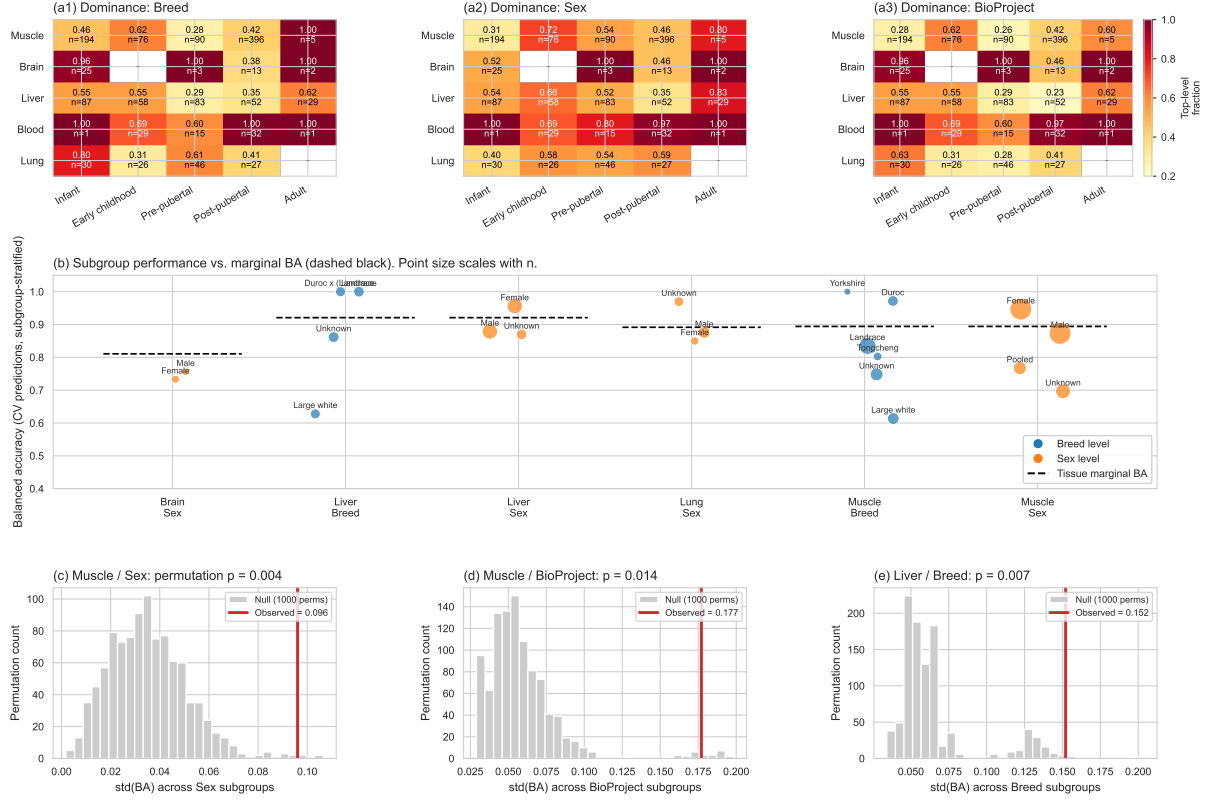


Supplementary Figure 7: Cross-species developmental conservation extended to brain and liver using the Cardoso-Moreira *et al.* 2019 human developmental atlas (E-MTAB-6814). Same pipeline as the original muscle analysis (strict one-to-one orthologs verified against Ensembl Compara; $\log_2$ fold change Infant vs Adult in both species; $\text{FDR} < 0.10$, $|\log_2 \text{FC}| > 0.5$ in both species as the “strict” filter; a pig-anchored variant using pig-only filtering is reported alongside, since the small per-tissue Cardoso-Moreira postnatal cohorts give underpowered human FDR). **Top row**, scatter plots of pig vs human $\log_2$ fold change for brain (left) and liver (right), with the strict-filter Pearson r and 95% bootstrap CI shown in each panel; lung is omitted because the Cardoso-Moreira human RPKM atlas does not contain lung samples. **Bottom**, forest plot summarising Pearson $r \pm 95\%$ CI across all four cross-species comparisons (Muscle from Schaiter *et al.* 2024 [previously published, $r = 0.69$, $n = 66$], Brain strict [$r = 0.26$, $n = 302$], Brain pig-anchored [$r = 0.19$, $n = 2,418$], Liver pig-anchored [$r = 0.25$, $n = 941$]). All correlations remain significantly positive; the brain strict result (81.5% directional concordance) is qualitatively compatible with the muscle finding, while liver requires a larger human postnatal cohort for a strict bidirectional test.

Supplementary Table 1: Cross-species developmental conservation across tissues. Per-tissue Pearson correlation of $\log_2$ fold change (Infant vs Adult) between pig (PigGTEx v0) and human (Cardoso-Moreira *et al.* 2019 unless otherwise noted) on strict one-to-one Ensembl Compara orthologs (the validated ortholog table is [review/analyses/results/pig_human_one_to_one_orthologs.csv](#); $n_{\text{orthologs}} = 8,277$). “Strict” uses $\text{FDR} < 0.10$, $|\log_2 \text{FC}| > 0.5$ in both species; “Pig-anchored” applies the filter in pig only (accommodating the small per-tissue postnatal n in the Cardoso-Moreira atlas). CI = 95% bootstrap percentile confidence interval (1,000 iterations). Per-tissue Pearson p -values are nominal; BH-FDR was applied gene-wise within each tissue but not across the panel of tissue-level tests. Lung is excluded because the Cardoso-Moreira human RPKM matrix does not include lung samples. Inspection of the brain strict-filter gene list revealed that several of the top infant-to-adult upregulated transcripts are canonical skeletal-muscle markers (e.g. *TNNC2*, *SMPX*, *MYBPHL*, *CHRNA1*, *LRRC2*), suggesting variable muscle-tissue contamination in a subset of pig brain biopsies; the brain strict r therefore partly reflects cross-species concordance of muscle developmental dynamics rather than brain-specific conservation. The liver pig-anchored directional concordance (47.3%) is below the 50% chance baseline, meaning the pig-anchored liver analysis provides no evidence of cross-species concordance.

Tissue	Filter	n_{genes}	Pearson r	95% CI	p -value	% concordant	Human source
Muscle	strict	66	0.693	0.54–0.80	1.1×10^{-10}	97.0	Schaiter 2024
Brain	strict	272	0.286	0.17–0.41	1.6×10^{-6}	85.3	Cardoso-Moreira 2019
Brain	pig-anchored	2,193	0.205	0.16–0.25	2.7×10^{-22}	38.5	Cardoso-Moreira 2019
Liver	strict	0	—	—	—	—	Cardoso-Moreira 2019
Liver	pig-anchored	857	0.213	0.14–0.29	3.0×10^{-10}	47.3	Cardoso-Moreira 2019
Lung	N/A	N/A	N/A	N/A	N/A	N/A	not in Cardoso-Moreira atlas

Figure S8 - Within-stage subgroup heterogeneity and its effect on per-tissue classifier performance



Supplementary Figure 8: Within-stage demographic heterogeneity and subgroup-sensitivity of the per-tissue developmental classifiers. **Top row**, sample-count heatmaps of (tissue $\times$ stage $\times$ subgroup) for the three stratifications: Breed, Sex, and BioProject. **Middle row**, box+strip plots of per-stratum balanced accuracy (BA) per tissue (subgroups with $n \geq 10$). Marginal BA is shown as a dashed line for each tissue. **Bottom panel**, permutation null distributions (1,000 within-stage label shuffles preserving each subgroup's stage composition) versus observed std(BA) for one representative tissue. Within-stage label permutation rejects the chance null for muscle/Sex ($p = 0.005$), muscle/BioProject ($p = 0.014$), and liver/Breed ($p = 0.008$); Brain/Sex, Liver/Sex, and Lung/Sex are consistent with chance ($p \geq 0.23$). Per-subgroup BA in muscle spans 0.61 (Large white, $n = 68$) to 1.00 (Yorkshire, $n = 12$). No subgroup with $n \geq 50$ falls below 0.70 for muscle or liver. Full per-subgroup statistics are in `review/analyses/results/heterogeneity_subgroup_performance.csv` and `.../heterogeneity_permutation_pvalues.csv`; per-stage demographic counts are in `.../heterogeneity_stratification.csv` (also Supplementary Table S9).

Supplementary Tables

Supplementary Table 2: Train versus test performance comparison. Both train and test (pooled cross-validated held-out predictions) performance are reported for all tissues. The gap between train and test metrics indicates the degree of overfitting; regularisation and early stopping were used to minimise this gap.

Tissue	Set	Scheme	n	Balanced Acc.	F1-macro	F1-weighted
Muscle	Train	4-class	761	0.968	0.974	0.984
Muscle	Test	4-class	761	0.894	0.899	0.939
Brain	Train	2-class	43	0.915	0.922	0.930
Brain	Test	2-class	43	0.811	0.801	0.816
Liver	Train	4-class	309	0.985	0.987	0.987
Liver	Test	4-class	309	0.921	0.918	0.919
Blood	Train	2-class	78	1.000	1.000	1.000
Blood	Test	2-class	78	0.959	0.960	0.961
Lung	Train	2-class	129	0.948	0.942	0.961
Lung	Test	2-class	129	0.892	0.876	0.916

Supplementary Table 3: Per-class performance metrics by tissue. Precision, recall, and F1 scores for each developmental stage within each tissue. n indicates the total number of samples per class (pooled across cross-validation folds).

Tissue	Stage	n	Precision	Recall	F1
Muscle	infant	194	0.939	0.959	0.949
Muscle	early childhood	76	0.838	0.816	0.827
Muscle	pre-pubertal	90	0.871	0.822	0.846
Muscle	post-pubertal/adult	401	0.973	0.980	0.976
Brain	pre-pubertal (<150d)	28	0.885	0.821	0.852
Brain	post-pubertal ($\geq$ 150d)	15	0.706	0.800	0.750
Liver	infant	87	0.965	0.954	0.960
Liver	early childhood	58	0.887	0.948	0.917
Liver	pre-pubertal	83	0.910	0.855	0.882
Liver	post-pubertal/adult	81	0.904	0.926	0.915
Blood	pre-pubertal (<150d)	45	0.957	0.978	0.967
Blood	post-pubertal ($\geq$ 150d)	33	0.969	0.939	0.954
Lung	pre-pubertal (<150d)	102	0.960	0.931	0.945
Lung	post-pubertal ($\geq$ 150d)	27	0.767	0.852	0.807

Supplementary Table 4: Functional annotations of cross-species developmental markers. Annotations compiled from UniProt, GeneCards, and literature review. FC = $\log_2$ fold change (adult/infant); positive values indicate higher expression in adults, negative values indicate higher expression in infants. Pig FC from pig GTEx data (GEO: GSE257558); Human FC from Schaiter *et al.* (2024). Genes ordered by model importance within each functional module. “—” indicates the gene did not pass cross-species criteria (FDR < 0.10, $|\log_2 \text{FC}| > 0.5$) but is included based on its known biological role.

Gene	Developmental Function	Pig FC	Human FC
<i>Module 1: Neuromuscular Maturation and Growth Signalling</i>			
<i>ACHE</i>	Functions as the key enzyme at the neuromuscular junction and as a structural adhesion molecule guiding neuronal growth during development[1].	−3.34	−0.53
<i>NRK</i>	Embryonic Ste20-type kinase that promotes actin polymerisation via cofilin phosphorylation and activates JNK signalling during skeletal muscle formation; expression restricted to embryonic and early postnatal development[2].	−8.56	−2.45
<i>KREMEN1</i>	Negative regulator of Wnt/ β -catenin signalling, partnering with Dkk1 to control cell survival and ensure proper patterning during postnatal development[3, 4].	−1.69	−1.43
<i>IGF2BP2</i>	RNA-binding protein that promotes myoblast proliferation via translational control of growth-related transcripts (c-Myc, Igflr, Sp1); postnatal downregulation required for terminal myogenic differentiation[5].	−2.19	−2.03
<i>SORCS2</i>	VPS10 domain-containing receptor that binds programulin to regulate motor neuron development during neuromuscular maturation[6].	−4.67	−0.95
<i>ETV5</i>	FGF-responsive PEA3-subfamily ETS transcription factor that promotes stem/progenitor cell self-renewal and myogenic differentiation in cooperation with MEF2; highly expressed during active postnatal myogenesis[7].	−2.80	−2.24
<i>IGF2BP3</i>	Oncofetal RNA-binding protein of the same family as IGF2BP2; expressed during embryonic development and silenced postnatally, consistent with its role in fetal growth programs.	−4.85	−1.23
<i>CCDC8</i>	Core component of the 3M complex (with CUL7 and OBSL1) that regulates microtubule dynamics, genome integrity, and growth hormone signalling during rapid postnatal growth; consistent with severe growth restriction in 3-M syndrome[8].	−4.44	−3.14
<i>Module 2: Structural Refinement and ECM Remodelling</i>			
<i>TCAF1</i>	Regulates TRPM8 ion channel trafficking and Ca^{2+} signalling; may support calcium-dependent processes during early postnatal muscle growth and metabolic maturation[9].	−4.72	−1.02
<i>FGFRL1</i>	Required for slow muscle fibre development; loss of function leads to diaphragm defects due to impaired slow fibre specification[10].	−3.03	−2.10
<i>WDR1</i>	Cofactor of cofilin-mediated actin filament disassembly, essential for sarcomere assembly and myofibril organisation during postnatal muscle growth; higher expression in immature muscle reflects intensive cytoskeletal remodelling during myofibrillogenesis[11].	−1.69	−0.52
<i>DLK1</i>	Crucial regulator of the myogenic program that inhibits myoblast proliferation while promoting differentiation; high infant expression declines as the muscle fiber population matures[12, 13].	−9.15	−2.88
<i>FSCN1</i>	Actin-bundling protein essential for formation of cellular protrusions during cell migration and tissue morphogenesis; highly expressed in embryonic somite-derived tissues, with postnatal downregulation as migratory programs complete[14].	−4.28	−1.95
<i>POSTN</i>	Secreted matricellular protein that mediates ECM remodelling, collagen fibrillogenesis, and connective tissue maturation during postnatal myofiber development; expression declines as tissue architecture stabilises[15].	−7.97	−2.93
<i>FREM2</i>	Basement membrane extracellular matrix protein of the FRAS/FREM complex maintaining epithelial–mesenchymal adhesion during organogenesis; postnatal downregulation reflects transition to collagen VII-dominated mature basement membranes[16].	−4.58	−1.05
<i>ADAMTS2</i>	Procollagen N-proteinase that cleaves propeptides from fibrillar procollagens to initiate collagen fibrillogenesis; high infant expression reflects intensive ECM biosynthesis during rapid postnatal growth[17].	−4.25	−1.83
<i>COL11A2</i>	Component of type XI collagen regulating collagen fibril organisation in cartilage and developing connective tissue; high infant expression reflects active skeletal morphogenesis and ECM assembly[18].	−8.35	−0.70
<i>TRIM54</i>	Muscle-specific E3 ubiquitin ligase (MuRF3) involved in sarcomere Z-disc assembly; expression increases postnatally as the mature contractile apparatus is established.	—	—
<i>Module 3: Metabolic Transition</i>			
<i>ME1</i>	Malic enzyme 1, catalysing oxidative decarboxylation of malate to pyruvate; a key source of NADPH for lipid biosynthesis. Expression increases during development, reflecting metabolic maturation.	2.59	0.64
<i>MAOB</i>	Mitochondrial outer membrane enzyme catalysing oxidative deamination of monoamines; postnatal upregulation reflects developmental maturation of oxidative metabolism and the glycolytic-to-oxidative fiber-type shift.	2.55	0.80
<i>PLA2G4E</i>	Calcium-dependent N-acyltransferase synthesising N-acylphosphatidylethanolamines (endocannabinoid precursors); species-discordant expression may reflect divergent intramuscular lipid metabolism programs.	2.32	−1.25

Supplementary Table 5: Stage-by-stage expression trajectories of cross-species developmental markers. Mean TPM values across porcine developmental stages for genes with conserved pig-human developmental patterns. Stage boundaries: infant (0–20 d), early childhood (21–59 d), pre-pubertal (60–149 d), post-pubertal (150–365 d), adult (>365 d). “Peak Change” indicates the developmental transition with the largest fold change. Genes are grouped by functional module and ordered by model importance within each module.

Gene	infant	early	pre-pub	post-pub	adult	Peak Change
<i>Module 1: Neuromuscular Maturation and Growth Signalling</i>						
<i>ACHE</i>	17.5	7.6	3.4	1.3	1.7	infant→early
<i>NRK</i>	128.2	4.2	8.1	1.3	0.3	infant→early
<i>KREMEN1</i>	30.3	13.2	9.4	5.2	9.4	infant→early
<i>IGF2BP2</i>	11.3	1.5	1.9	1.0	2.5	infant→early
<i>SORCS2</i>	14.4	3.0	2.2	0.6	0.6	infant→early
<i>ETV5</i>	34.2	16.7	7.5	5.5	4.9	early→pre-pub
<i>IGF2BP3</i>	21.8	1.7	2.1	0.4	0.7	infant→early
<i>CCDC8</i>	177.4	28.2	25.4	13.4	8.2	infant→early
<i>Module 2: Structural Refinement and ECM Remodelling</i>						
<i>TCAF1</i>	78.9	8.3	8.1	3.8	3.0	infant→early
<i>FGFRL1</i>	77.4	15.6	14.2	4.3	9.5	infant→early
<i>WDR1</i>	202.9	125.6	95.0	51.9	63.1	pre-pub→post-pub
<i>DLK1</i>	338.8	3.9	6.4	0.7	0.6	infant→early
<i>FSCN1</i>	123.7	32.5	18.9	6.2	6.4	infant→early
<i>POSTN</i>	602.4	44.1	54.3	2.5	2.4	pre-pub→post-pub
<i>FREM2</i>	33.1	4.4	5.5	1.5	1.4	infant→early
<i>ADAMTS2</i>	51.1	16.6	16.0	6.4	2.7	infant→early
<i>COL11A2</i>	44.6	1.5	1.0	0.3	0.1	infant→early
<i>TRIM54</i>	78.3	351.1	149.4	377.9	144.4	infant→early
<i>Module 3: Metabolic Transition</i>						
<i>ME1</i>	2.8	4.0	3.9	6.9	16.9	post-pub→adult
<i>MAOB</i>	5.7	24.2	10.8	15.7	33.4	infant→early
<i>PLA2G4E</i>	0.1	1.4	0.7	0.6	0.5	infant→early

Supplementary Table 6: Within-project fold-change validation across developmental contrasts. Developmental fold-changes ($\log_2$ scale) were computed within single-project muscle cohorts—each spanning four developmental stages under uniform laboratory protocols—and correlated with full-dataset estimates across three stage comparisons (all using infant as baseline). Pooled within-project fold-changes were computed by averaging across both projects before correlating with full-dataset estimates. “All genes” includes all 31,908 expressed genes; “Cross-species” includes the 66 genes passing pre-specified criteria ($\text{FDR} < 0.10$, $|\log_2 \text{FC}| > 0.5$) in both pig and human. CI = 95% bootstrap confidence interval (1,000 iterations). Per-project results are available in the analysis repository. Full-dataset sample sizes: infant $n = 194$, early childhood $n = 76$, pre-pubertal $n = 90$, post-pubertal $n = 396$ (excludes 5 adult samples merged into post-pubertal/adult in the 4-class scheme).

Comparison	Gene set	Full dataset		Within-project		Pooled r [95% CI]	p -value
		n_{Inf}	n_{later}	n_{Inf}	n_{later}		
infant vs. early childhood	All genes	194	76	106	12	0.669 [0.66–0.68]	$< 10^{-300}$
	Cross-species					0.761	1.3×10^{-13}
infant vs. pre-pubertal	All genes	194	90	106	27	0.806 [0.80–0.82]	$< 10^{-300}$
	Cross-species					0.965	6.9×10^{-39}
infant vs. post-pubertal	All genes	194	396	106	10	0.799 [0.79–0.81]	$< 10^{-300}$
	Cross-species					0.978	5.6×10^{-45}

Supplementary Table 7: Variance decomposition of PC1–PC5 with respect to developmental stage and BioProject across five tissues. For each tissue we projected the $\log_2(\text{TPM} + 1)$ values of the top 5,000 most-variable genes into a 10-component PCA. For each principal component we fit ordinary-least-squares models $\text{PC} \sim \text{Stage}$, $\text{PC} \sim \text{BioProject}$, and the combined $\text{PC} \sim \text{Stage} + \text{BioProject}$, and report the coefficient of determination (R^2). n_{samples} is the number of samples per tissue available after merging metadata; n_{projects} is the number of distinct BioProjects represented. BioProject explains the larger absolute variance share in every tissue because most PigGTEx projects sample a single narrow age window (Stage and BioProject are partially confounded by design); however, stage alone still explains a substantial fraction of PC1 variance in muscle, brain, lung, and blood, and is recovered at later PCs in liver ($\text{PC5 } R^2_{\text{stage}} = 0.29$).

Tissue	PC	R^2_{stage}	$R^2_{\text{bioproject}}$	R^2_{combined}	n_{projects}	n_{samples}
Muscle	PC1	0.403	0.803	0.884	56	761
Muscle	PC2	0.242	0.750	0.798	56	761
Muscle	PC3	0.156	0.862	0.884	56	761
Muscle	PC4	0.242	0.718	0.797	56	761
Muscle	PC5	0.038	0.846	0.848	56	761
Brain	PC1	0.648	0.954	0.956	6	43
Brain	PC2	0.308	0.922	0.922	6	43
Brain	PC3	0.803	0.991	0.994	6	43
Brain	PC4	0.163	0.923	0.943	6	43
Brain	PC5	0.136	0.478	0.504	6	43
Liver	PC1	0.016	0.935	0.936	34	309
Liver	PC2	0.167	0.962	0.963	34	309
Liver	PC3	0.216	0.920	0.920	34	309
Liver	PC4	0.110	0.869	0.880	34	309
Liver	PC5	0.292	0.641	0.778	34	309
Lung	PC1	0.333	0.881	0.916	20	129
Lung	PC2	0.563	0.895	0.914	20	129
Lung	PC3	0.218	0.813	0.842	20	129
Lung	PC4	0.077	0.752	0.800	20	129
Lung	PC5	0.119	0.904	0.904	20	129
Blood	PC1	0.545	0.806	0.817	7	78
Blood	PC2	0.370	0.954	0.960	7	78
Blood	PC3	0.344	0.727	0.746	7	78
Blood	PC4	0.128	0.794	0.814	7	78
Blood	PC5	0.327	0.914	0.930	7	78

Supplementary Table 8: Top 20 developmental-stage features per tissue. Genes ranked by aggregate Light-GBM importance across the K–1 binary classifiers within the calibrated ordinal model for each tissue. Functional descriptions are drawn from mygene.info (<https://mygene.info/>) and UniProt (<https://rest.uniprot.org/>); placeholder entries equal the gene symbol indicate that no curated description was returned by either source. “Peak stage” is the developmental stage with the highest mean $\log_2(\text{TPM}+1)$; “FC” is the peak minus trough in the same units.

Tissue	Rank	Gene	Function	Peak stage	FC
Muscle	1	<i>SORCS2</i>	VPS10 domain-containing receptor SorCS2.	Infant	2.87
	2	<i>ARHGAP36</i>	GTPase activator for the Rho-type GTPases by converting them to an inactive GDP-bound state.	Infant	4.76
	3	<i>FGFRL1</i>	Has a negative effect on cell proliferation.	Infant	3.98
	4	<i>IGSF1</i>	Immunoglobulin superfamily member 1.	Infant	4.04
	5	<i>LOC100519808</i>	TRPM8 channel associated factor 1.	Infant	3.90
	6	<i>ACTC1</i>	Actin alpha cardiac muscle 1.	Infant	10.14
	7	<i>LOC102158612</i>	LOC102158612	Post-pubertal	4.89
	8	<i>PPP1R14C</i>	Inhibitor of PPP1CA.	Early childhood	4.61
	9	<i>ACHE</i>	Carboxylic ester hydrolase.	Infant	2.67
	10	<i>LOC110257362</i>	LOC110257362	Infant	3.17
	11	<i>TRDN</i>	Contributes to the regulation of luminal Ca ²⁺ release via the sarcoplasmic reticulum calcium release channels RYR1 and RYR2, a key step in triggering skeletal and heart muscle contraction. Required for normal organization of the triad junction, where T-tubules and the.	Post-pubertal	2.09
	12	<i>DLK1</i>	Delta like non-canonical Notch ligand 1.	Infant	6.34
	13	<i>ENSSSCG00000042876</i>	ENSSSCG00000042876	Post-pubertal	1.43
	14	<i>SRPX</i>	Sushi repeat containing protein X-linked.	Infant	4.53
	15	<i>WDR1</i>	WD repeat-containing protein 1.	Infant	2.18
	16	<i>NRK</i>	Nik related kinase.	Infant	5.38
	17	<i>SLC22A16</i>	Facilitative organic cation transporter that mediates the transport of carnitine as well as the polyamine spermidine. Mediates the partially Na(+)-dependent bidirectional transport of carnitine.	Post-pubertal	4.18
	18	<i>ENSSSCG00000044729</i>	ENSSSCG00000044729	Post-pubertal	1.25
	19	<i>ETV4</i>	ETS variant transcription factor 4.	Infant	2.93
	20	<i>GKAP1</i>	G kinase anchoring protein 1.	Adult	1.38
Brain	1	<i>ENSSSCG00000046117</i>	ENSSSCG00000046117	Post-pubertal	0.60
	2	<i>CCDC9B</i>	Coiled-coil domain containing 9B.	Post-pubertal	1.93
	3	<i>SAMD5</i>	Sterile alpha motif domain-containing protein 5.	Post-pubertal	1.09
	4	<i>ENSSSCG00000038527</i>	Uncharacterized protein.	Adult	0.72
	5	<i>ENSSSCG00000045405</i>	ENSSSCG00000045405	Post-pubertal	0.53
	6	<i>HMGCS2</i>	Catalyzes the condensation of acetyl-CoA with acetoacetyl-CoA to form HMG-CoA.	Infant	0.22
	7	<i>LATS1</i>	non-specific serine/threonine protein kinase.	Post-pubertal	1.51
	8	<i>GLIS3</i>	GLIS family zinc finger 3.	Post-pubertal	1.33
	9	<i>LOC100158196</i>	Electron carrier protein. The oxidized form of the cytochrome c heme group can accept an electron from the heme group of the cytochrome c1 subunit of cytochrome reductase.	Infant	1.68
	10	<i>NOTUM</i>	Notum, palmitoleoyl-protein carboxylesterase.	Pre-pubertal	1.50

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Tissue	Rank	Gene	Function	Peak stage	FC
	11	<i>SLC6A20</i>	Transporter.	Post-pubertal	3.44
	12	<i>FIBCD1</i>	Fibrinogen C domain containing 1.	Infant	4.02
	13	<i>LAMA1</i>	Anti-angiogenic and anti-tumor peptide that inhibits endothelial cell migration, collagen-induced endothelial tube morphogenesis and blood vessel growth in the chorioallantoic membrane. Blocks endothelial cell adhesion to fibronectin and type I collagen.	Post-pubertal	1.34
	14	<i>LRRC71</i>	Leucine rich repeat containing 71.	Adult	1.04
	15	<i>SASH1</i>	Is a positive regulator of NF-kappa-B signaling downstream of TLR4 activation. It acts as a scaffold molecule to assemble a molecular complex that includes TRAF6, MAP3K7, CHUK and IKKBK, thereby facilitating NF-kappa-B signaling activation.	Post-pubertal	2.54
	16	<i>LOC100516957</i>	Adhesion G protein-coupled receptor E2.	Pre-pubertal	0.13
	17	<i>CCDC170</i>	Coiled-coil domain containing 170.	Adult	2.76
	18	<i>CEP162</i>	Centrosomal protein of 162 kDa.	Post-pubertal	1.49
	19	<i>ACAP3</i>	GTPase-activating protein for the ADP ribosylation factor family.	Infant	0.53
	20	<i>ENSSSCG00000033041</i>	ENSSSCG00000033041	Adult	2.43
Liver	1	<i>SATB2</i>	DNA-binding protein SATB.	Post-pubertal	0.47
	2	<i>CDH7</i>	Cadherins are calcium-dependent cell adhesion proteins.	Post-pubertal	0.99
	3	<i>ALDH18A1</i>	Bifunctional enzyme that converts glutamate to glutamate 5-semialdehyde, an intermediate in the biosynthesis of proline, ornithine and arginine.	Infant	3.42
	4	<i>ENSSSCG00000036118</i>	ENSSSCG00000036118	Early childhood	2.12
	5	<i>ENSSSCG00000050649</i>	ENSSSCG00000050649	Infant	5.87
	6	<i>COL1A1</i>	Collagen type I alpha 1 chain.	Infant	1.90
	7	<i>EIF3G</i>	RNA-binding component of the eukaryotic translation initiation factor 3 (eIF-3) complex, which is required for several steps in the initiation of protein synthesis. The eIF-3 complex associates with the 40S ribosome and facilitates the recruitment of eIF-1, eIF-1A,.	Early childhood	0.96
	8	<i>CD320</i>	CD320 molecule.	Infant	2.03
	9	<i>ACSF2</i>	Acyl-CoA synthases catalyze the initial reaction in fatty acid metabolism, by forming a thioester with CoA. Has some preference toward medium-chain substrates.	Infant	1.92
	10	<i>LOC100519366</i>	Cytochrome c oxidase subunit 6B1.	Post-pubertal	1.79
	11	<i>FAM169B</i>	Protein FAM169B.	Early childhood	2.63
	12	<i>DEK</i>	Involved in chromatin organization.	Adult	1.10
	13	<i>ENSSSCG00000033368</i>	ENSSSCG00000033368	Infant	1.28
	14	<i>TK1</i>	Thymidine kinase.	Infant	2.59
	15	<i>SYN3</i>	Synapsin III.	Post-pubertal	0.58
	16	<i>RPL35A</i>	Component of the large ribosomal subunit. The ribosome is a large ribonucleoprotein complex responsible for the synthesis of proteins in the cell.	Early childhood	1.42
	17	<i>PTCHD1</i>	Required for the development and function of the thalamic reticular nucleus (TRN), a part of the thalamus that is critical for thalamocortical transmission, generation of sleep rhythms, sensorimotor processing and attention. Can bind cholesterol in vitro.	Post-pubertal	0.44

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Tissue	Rank	Gene	Function	Peak stage	FC
	18	<i>RARRES1</i>	Retinoic acid receptor responder 1.	Post-pubertal	2.11
	19	<i>GARNL3</i>	GTPase-activating Rap/Ran-GAP domain-like protein 3.	Post-pubertal	2.44
	20	<i>RAI2</i>	Retinoic acid induced 2.	Early childhood	1.20
Lung	1	<i>CCN5</i>	May play an important role in modulating bone turnover. Promotes the adhesion of osteoblast cells and inhibits the binding of fibrinogen to integrin receptors.	Post-pubertal	2.71
	2	<i>L1CAM</i>	Neural cell adhesion molecule involved in the dynamics of cell adhesion and in the generation of transmembrane signals at tyrosine kinase receptors. During brain development, critical in multiple processes, including neuronal migration, axonal growth and fasciculation, and.	Post-pubertal	1.74
	3	<i>GZMM</i>	Cleaves peptide substrates after methionine, leucine, and norleucine. Physiological substrates include EZR, alpha-tubulins and the apoptosis inhibitor BIRC5/Survivin.	Post-pubertal	2.44
	4	<i>HMGN1</i>	High mobility group nucleosome binding domain 1.	Infant	0.98
	5	<i>IGF2BP2</i>	RNA-binding factor that recruits target transcripts to cytoplasmic protein-RNA complexes (mRNPs). This transcript 'caging' into mRNPs allows mRNA transport and transient storage.	Infant	1.57
	6	<i>DSC2</i>	Desmocollin 2.	Infant	1.21
	7	<i>LOC102158691</i>	Immunoglobulin V-set domain-containing protein.	Post-pubertal	4.04
	8	<i>ADAM22</i>	ADAM metallopeptidase domain 22.	Post-pubertal	0.62
	9	<i>IGSF1</i>	Immunoglobulin superfamily member 1.	Infant	0.80
	10	<i>ENSSSCG00000012668</i>	Putative odorant or sperm cell receptor.	Post-pubertal	0.02
	11	<i>ENSSSCG00000041871</i>	ENSSSCG00000041871	Post-pubertal	0.01
	12	<i>LOC100125545</i>	LOC100125545	Early childhood	0.36
	13	<i>LOC110257692</i>	Putative odorant or sperm cell receptor.	Infant	0.06
	14	<i>ARHGAP36</i>	GTPase activator for the Rho-type GTPases by converting them to an inactive GDP-bound state.	Infant	0.18
	15	<i>LOC100511059</i>	Putative odorant or sperm cell receptor.	Early childhood	0.03
	16	<i>LOC100513516</i>	Putative odorant or sperm cell receptor.	Early childhood	0.03
	17	<i>OR13H1</i>	Putative odorant or sperm cell receptor.	Early childhood	0.04
	18	<i>LOC100514087</i>	Putative odorant or sperm cell receptor.	Early childhood	0.03
	19	<i>ENSSSCG00000048571</i>	ENSSSCG00000048571	Early childhood	0.05
	20	<i>ENSSSCG00000035001</i>	ENSSSCG00000035001	Pre-pubertal	0.02
Blood	1	<i>RPLP1</i>	Plays an important role in the elongation step of protein synthesis.	Infant	3.14
	2	<i>ENSSSCG00000033734</i>	ENSSSCG00000033734	Early childhood	8.93
	3	<i>ENSSSCG00000042133</i>	ENSSSCG00000042133	Adult	3.05
	4	<i>USF2</i>	Transcription factor that binds to a symmetrical DNA sequence (E-boxes) (5'-CACGTG-3') that is found in a variety of viral and cellular promoters.	Infant	3.10
	5	<i>BOLA2B</i>	Acts as a cytosolic iron-sulfur (Fe-S) cluster assembly factor that facilitates [2Fe-2S] cluster insertion into a subset of cytosolic proteins. Acts together with the monothiol glutaredoxin GLRX3.	Infant	3.35
	6	<i>TMEM158</i>	Transmembrane protein 158.	Post-pubertal	1.95
	7	<i>CCR6</i>	C-C motif chemokine receptor 6.	Post-pubertal	2.33
	8	<i>BPIFB4</i>	BPI fold containing family B member 4.	Post-pubertal	0.86

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Tissue	Rank	Gene	Function	Peak stage	FC
	9	<i>ENSSSCG00000045920</i>	ENSSSCG00000045920	Infant	3.56
	10	<i>PKP2</i>	Plakophilin-2.	Adult	1.39
	11	<i>XPO7</i>	Exportin 7.	Infant	4.71
	12	<i>ENSSSCG00000051365</i>	ENSSSCG00000051365	Adult	3.76
	13	<i>ENSSSCG00000045536</i>	ENSSSCG00000045536	Adult	4.10
	14	<i>TLL1</i>	Metalloendopeptidase.	Adult	0.27
	15	<i>NSMCE4A</i>	Component of the SMC5-SMC6 complex, that promotes sister chromatid alignment after DNA damage and facilitates double-stranded DNA breaks (DSBs) repair via homologous recombination between sister chromatids.	Post-pubertal	1.82
	16	<i>NT5M</i>	5',3'-nucleotidase, mitochondrial.	Post-pubertal	1.21
	17	<i>MIR9796</i>	microRNAs (miRNAs) are short (20-24 nt) non-coding RNAs that are involved in post-transcriptional regulation of gene expression in multicellular organisms by affecting both the stability and translation of mRNAs. miRNAs are transcribed by RNA polymerase II as part of capped and.	Post-pubertal	0.04
	18	<i>ENSSSCG00000031003</i>	Uncharacterized protein.	Post-pubertal	4.29
	19	<i>LOC100626120</i>	Ig-like domain-containing protein.	Post-pubertal	1.33
	20	<i>ENSSSCG00000034023</i>	ENSSSCG00000034023	Early childhood	0.55

Supplementary Table 9: Within-stage label permutation tests for subgroup-balanced-accuracy heterogeneity. For each (tissue $\times$ stratification) pair, the observed standard deviation of subgroup-level balanced accuracy (BA, restricted to subgroups with $n \geq 10$) is compared to a null distribution generated by 1,000 within-stage label shuffles (preserving each subgroup’s stage composition). p -values are one-sided (probability of observed-or-larger std under the null). Subgroups whose marginal balanced accuracy varies beyond chance are highlighted by significant p -values; non-significant rows indicate that subgroup-level performance variation is statistically indistinguishable from random label permutation given the available sample sizes. Full per-subgroup statistics are in `review/analyses/results/heterogeneity_subgroup_performance.csv`.

Tissue	Stratification	Observed std(BA)	Permutation p	Significant?
Muscle	Sex	0.096	0.005	yes
Muscle	BioProject	0.177	0.015	yes
Muscle	Breed	0.131	0.054	borderline
Liver	Breed	0.152	0.008	yes
Brain	Sex	0.012	1.000	no
Liver	Sex	0.039	0.232	no
Lung	Sex	0.051	0.480	no

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